

Question	Answer	Marks
1(a)	Rate of change of displacement	B1
1(b)	For object A: $s = ut + \frac{1}{2}at^2$ so $t = \sqrt{\frac{2 \times 10}{9.81}}$ =1.4	C1
	Then for object B: $s = ut + \frac{1}{2}at^2$ $h = -(3 \times 1.4) + (0.5 \times 9.81 \times 1.4^2)$	C1
	= 5.7 m	A1
	OR $h = -(3 \times 1.4) + 10$	(C1)
	= 5.7 m	(A1)
1(c)(i)	time taken (to reach the ground is) same	B1
	The <u>initial vertical</u> (component of the) velocity is the same (as in part (1b))	B1
1(c)(ii)	The (total) initial energy is greater (than in part (1b))	B1
	change in gravitational potential energy is same, so speed is greater	B1

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